

Design of a 32nm Ultra-Low Power 6T SRAM Cell Analyzing Emerging Technologies such as FinFET, TFET, and CNFET for energy-efficient applications

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Abstract—This work presents the design, simulation, and comparative analysis of a 6-transistor (6T) static random-access memory (SRAM) cell optimized for ultra-low-power operation. Emerging device technologies—Fin Field-Effect Transistor (FinFET), Tunnel Field-Effect Transistor (TFET), and Carbon Nanotube Field-Effect Transistor (CNFET)—are evaluated individually and in hybrid configurations to overcome the limitations of conventional CMOS SRAM cells. Device-level models are calibrated to ensure fair comparison, matching leakage currents and parasitic capacitances. Using NGSPICE and HSPICE simulations at subthreshold supply voltages (0.6–0.8 V), we measure static, dynamic, and total power consumption for each configuration. Results show that the FinFET-TFET hybrid achieves the lowest total power dissipation, combining the low leakage of TFETs with the high speed and channel control of FinFETs, making it suitable for energy-constrained applications such as IoT and implantable devices.

Index Terms—Ultra-low-power SRAM, FinFET, TFET, CNFET, hybrid technology, low-voltage design, power consumption.

I. INTRODUCTION

Six-transistor (6T) SRAM cells are fundamental in high-performance applications due to their simplicity and operational efficiency [1]. However, their energy consumption, particularly during read and write operations, remains a significant challenge in embedded and portable systems where energy efficiency is critical. Recently, emerging device technologies have been introduced to overcome these limitations. FinFETs, with their three-dimensional structure, provide superior electrostatic control and reduced leakage, enabling stable operation at lower voltages [1], [2]. CNFETs also show high carrier mobility and notable leakage reduction, allowing the design of SRAM cells with improved static and dynamic power performance [3]. TFETs, exploiting band-to-band tunneling, enable ultra-low voltage operation, and recent architectures such as disturb-free 10T cells [4] and hybrid 8T TFET-MOSFET designs [5] have demonstrated significant gains in static noise margin and power efficiency. This research focuses on the design of 6T SRAM cells using FinFET, TFET, and CNFET technologies, with the primary goal of reducing energy consumption. Through calibrated simulations in NGSPICE and

HSPICE, the 6T architecture is evaluated in terms of static and dynamic power, showing notable improvements in energy efficiency without compromising stability.

II. METHODOLOGY

The work focuses on the design and evaluation of 6-transistor (6T) SRAM cells at 32nm optimized to reduce energy consumption, using CMOS, FinFET, CNFET, and TFET technologies. An experimental methodology based on circuit simulations was used to directly compare the impact of each technology on consumption, while maintaining the basic functionality of the cell. The CMOS and FinFET models were taken from the literature with parameters for static and dynamic analysis [6], the CNFET was modeled according to the proposal from the University of Minnesota, which considers mobility, parasitic capacitances, and leakage current [7], and the TFET was implemented using a model based on TCAD simulations for ultra-low voltage operation [8]. Detailed structures and characteristics of each technology can be found in the cited references [6]–[8]

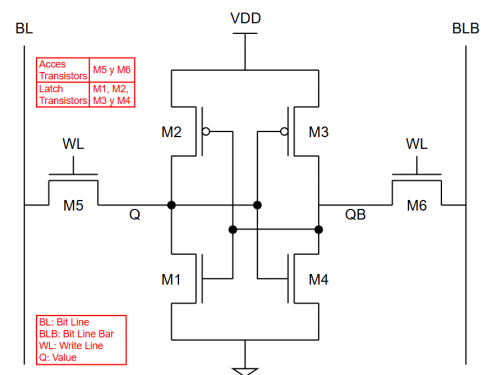


Fig. 1: Schematic of a 6T SRAM cell

The cell designs were created in Xschem, generating the netlists that were then simulated in HSPICE, allowing the extraction of consumption and stability metrics under subthreshold conditions (0.6 V to 0.8 V).

TABLE I: Proposed cases with the mentioned technologies

Case	Latch Transistors	Access Transistors
1	NMOS	NMOS
2	NFinFET	NFinFET
3	NCNFET	NCNFET
4	NTFET	NTFET
5	NCNFET	NTFET
6	NCNFET	NMOS
7	NMOS	NTFET
8	NFinFET	NCNFET
9	NFinFET	NMOS

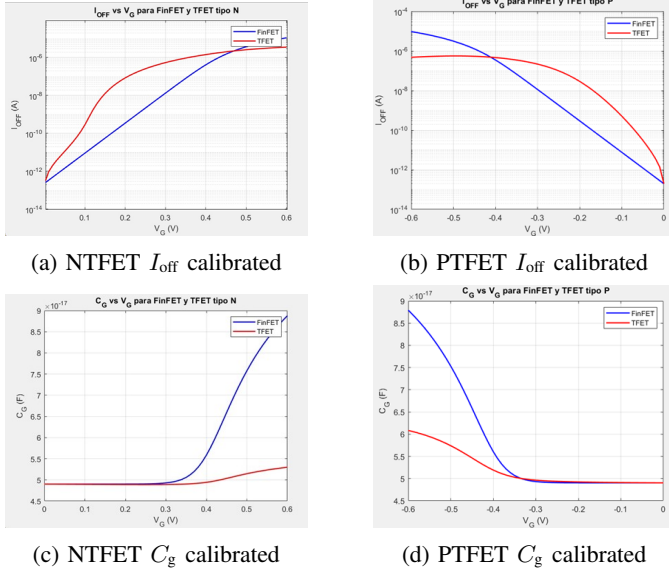


Fig. 2: TFET model calibration

The device models are calibrated primarily for leakage currents (I_{off}) and parasitic capacitances ($C_g = C_{gs} + C_{gd}$), using the Verilog-A TFET model. The calibration employs three key curves: CGS, CGD, and I_{off} , which are critical for modeling device behavior as seen in previous studies[9]. However, parameters such as ON/OFF resistors, leakage inductance, and voltage thresholds will be considered in subsequent studies to enhance model accuracy and provide a more comprehensive evaluation of power dissipation in SRAM cells.

III. RESULTS

By designing both the individual and hybrid cells, static, dynamic, and total consumption results were obtained.

TABLE II: Static, dynamic, and total power consumption

technology	VDD (V)	P_{sta} (W)	P_{dyn} (W)	P_{tot} (W)
CMOS	0.8	5.67E-10	2.70E-05	2.64E-07
CNFET	0.6	2.12E-10	4.53E-06	1.26E-08
FinFET	0.6	2.80E-12	2.79E-06	9.49E-09
TFET	0.6	4.19E-09	1.64E-06	2.66E-08
TFET-CMOS	0.6	1.87E-09	1.67E-06	5.89E-08
FinFET-TFET	0.6	1.69E-10	1.10E-06	6.11E-09
FinFET-CNFET	0.6	4.99E-11	3.25E-06	1.17E-08
FinFET-CMOS	0.6	4.81E-12	3.80E-06	4.46E-08
CNFET-TFET	0.6	3.04E-10	6.67E-06	3.29E-08
CNFET-CMOS	0.8	2.14E-10	2.86E-05	1.16E-07

The cell current was measured in an idle state, with static power calculated from current and supply voltage (0.6 V and 0.8 V). Dynamic consumption was evaluated by applying write and read transitions. To determine total consumption, a 60 ns interval was simulated, capturing both static and dynamic operations. This interval ensures scalability (repetitive switching behavior) and computational efficiency, providing a sufficient analysis of power consumption for the given low-voltage operation. Although it may be narrow for some IoT applications, it is adequate for the scope of this study.

IV. CONCLUSIONS

The evaluation of 6T SRAM cells using FinFET, TFET, and CNFET technologies confirmed that hybrid approaches, particularly the FinFET–TFET combination, can significantly reduce total power consumption while maintaining stability. These results reinforce the potential of emerging device technologies to improve energy efficiency in memory design.

However, this study was limited to a narrow voltage range and single-cell analysis, without exploring variability through corner simulations or the scalability of large arrays and peripheral circuits. Extending the analysis to include these aspects, along with updated benchmarking against recent device models, will provide a more comprehensive validation of hybrid SRAM designs for practical, ultra-low power applications.

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